

1



SMART INDIA
HACKATHON
2022



ग्रामीण विकास मंत्रालय
MINISTRY OF
RURAL DEVELOPMENT

Integrated GIS-based Digital Public Infrastructure for Land Governance

Transparent | Secure | Accessible | Inclusive

*“A Smarter Tomorrow
for Every Piece of Land”*

Team name : Last not Least

Team Members : Rajat Chakraborty (EE, Final Year) & Asmita Sarkar (EE, Final Year)

Institute : Tripur Institute of Technology

Problem Statement ID : SIH26014



2

The Problem We Are Solving



Fragmented
land records



Time-consuming
manual processes



No real-time
visibility into land
status



Multiple departments
and databases



Citizens face
disputes, fraud
and poor access



Lack of real-time
spatial information



Inconsistent and
outdated records



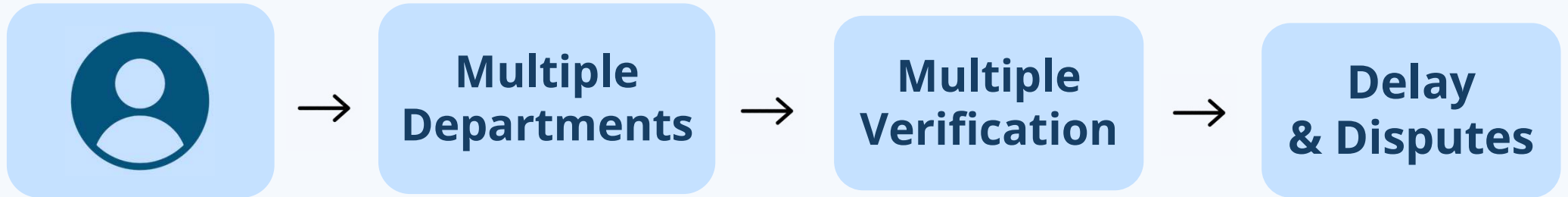
Outdated
Lands

***“ Land Information exists, but it is not sufficiently
connected, accessible & actionable ”***



3

Current System & Its Gaps



Existing Approach

Paper/Manual records

Separate databases

Multiple Verification

Limited GIS integration

Physical visit

Limited transparency



Problems

Slow & error-prone

Data fragmentation

Time Consuming

Poor spatial understanding

Inconvenient

Disputes/fraud risk



4

Our Proposed Solution



"One integrated platform connections citizen, land records, GIS data & authorities."



5

How the Platform Works

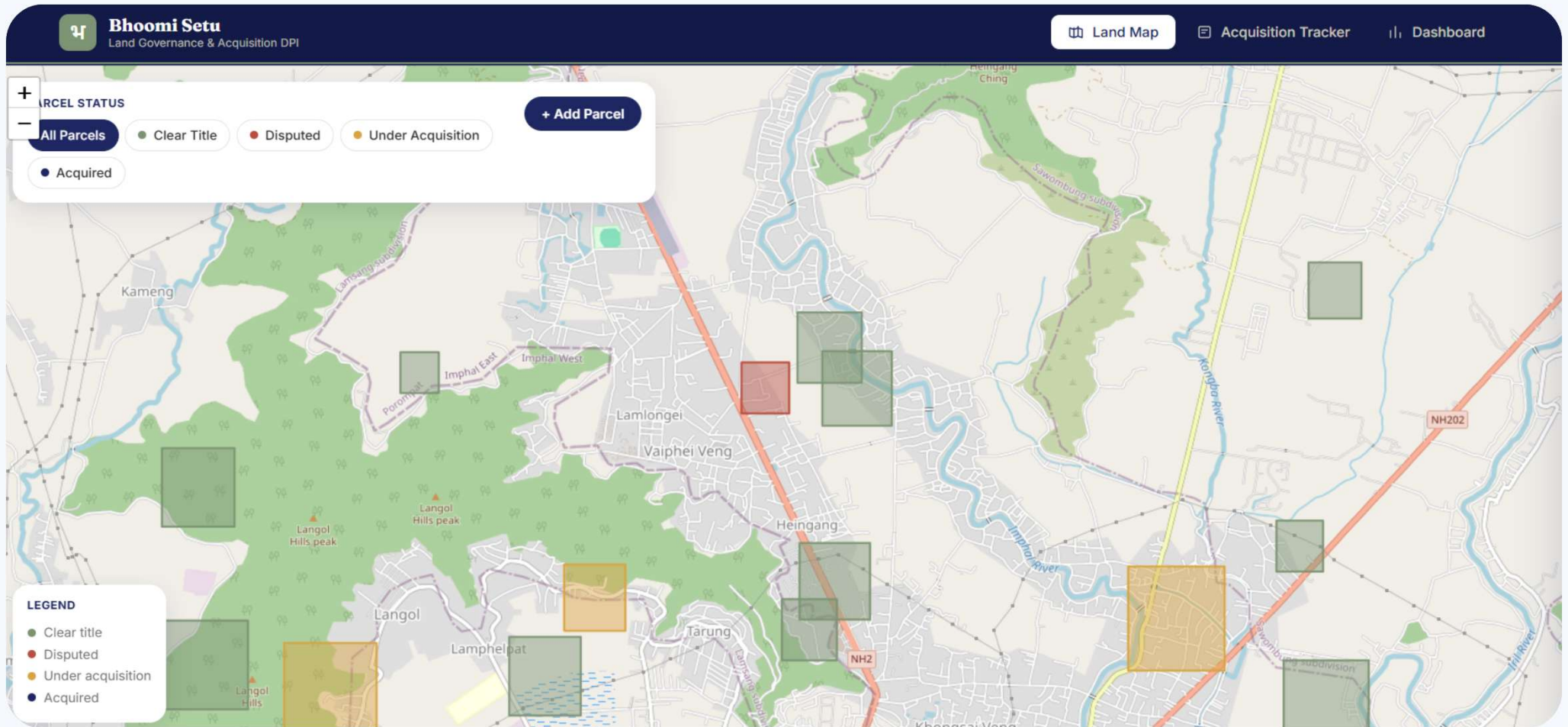


From search to settlement – All in one platform



6

GIS-Based Land Map

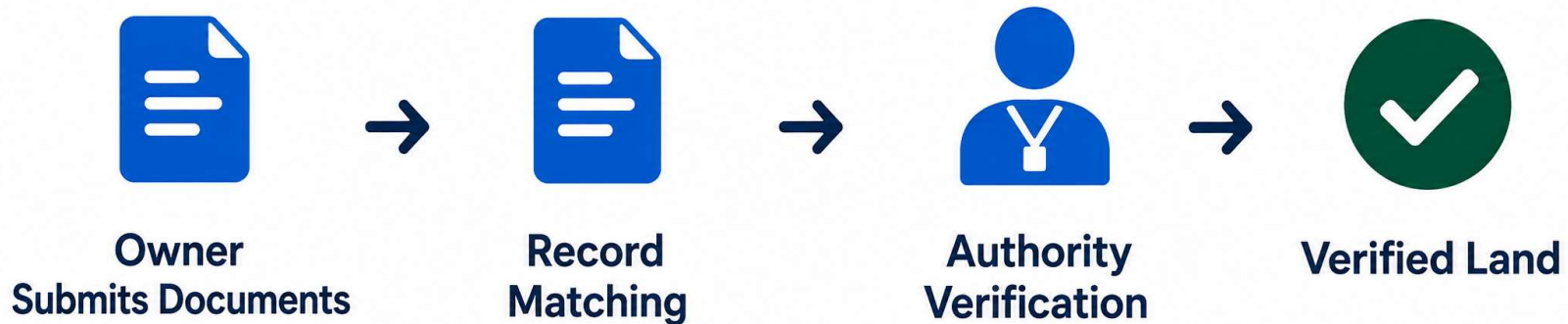
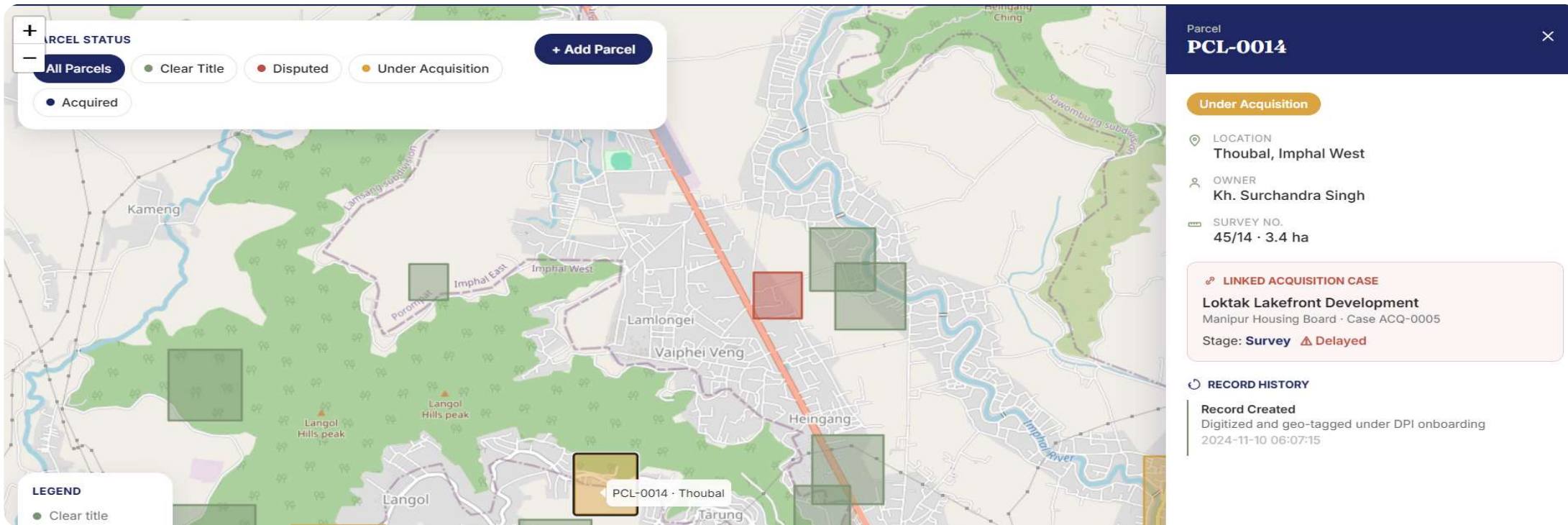


Every land parcel becomes a digitally identifiable spatial asset



6

GIS-Based Land Map



8

Secure Land Marketplace



Sell Land

- ✓ List your property
- ✓ Add land details
- ✓ Upload documents
- ✓ Get verified
- ✓ Reach genuine buyers

List Property



Buy Land

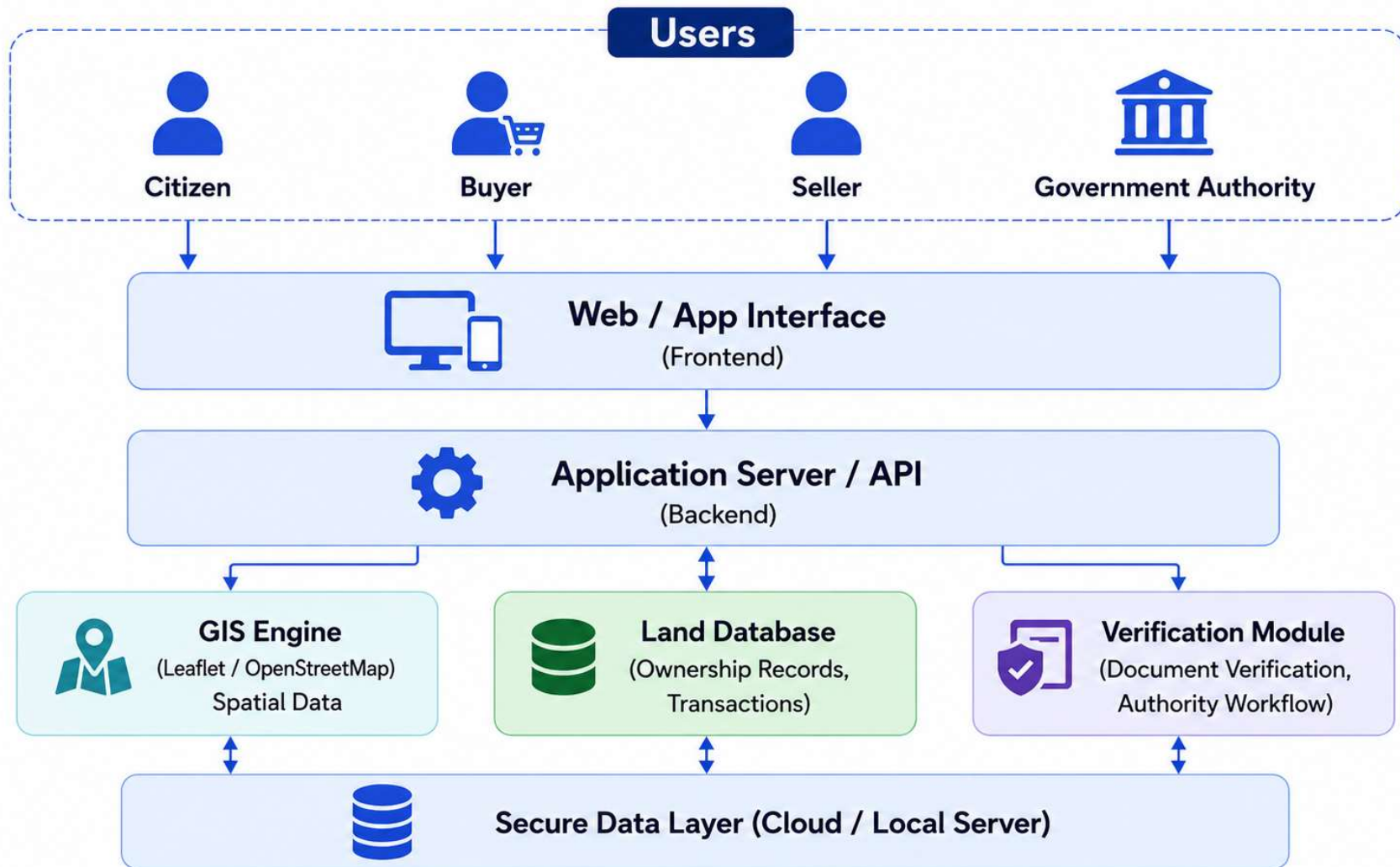
- ↑ Search on GIS map
- ↑ View verified details
- ↑ Check documents
- ↑ Submit purchase request
- ↑ Track status

Search Property

A transparent & secure marketplace for land transaction



System Architecture



Security, Transparency & Trust



10 Expected Impact



Processing Time



Transparency



Fraud Risk



Citizen Accessibility



Manual Paperwork



Data Accuracy

Beneficiaries



Citizens



Land Owners



Buyers



Sellers



Government
Authorities



Survey/Revenue
Departments



